



UPI QR Code Encoding Specification

Merchant Presented QRC



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About this Document

Purpose

The QR Code (QRC) enjoys a wide array of applications in many areas and is now booming in the payment industry. Currently, the merchant-presented QRC and the consumer-presented QRC are both common market practices. Several solutions including barcode, QRC with chip information, etc., are emerging on a world-wide scale.

To better assist Members and Partners outside of Mainland China in implementing UnionPay International (UPI) QRC products, this document is provided with a detailed description of the information contained in the merchant-presented QRC data.

Audience

This document is intended for APP providers, the merchant-presented QRC generators, etc..

Scope

This document includes:

- The encoding of the merchant-presented QRC
- The cardholder's scanning the QRC outside of Mainland China

This document excludes:

- The consumer-presented QRC use case
- QRC transactions inside Mainland China
- The message requirements, settlement requirements, APP and terminal requirements other than encoding.

The requirements for the QRC displayed by the Merchant, including its format and content, are described in the *EMV® QR Code Specification for Payment Systems (EMV QRCPS) Merchant-Presented Mode*.

References

The terms and conditions in this document have quoted the following publications. Any later versions of these publications will be applied in this document unless specified.

- *Technical Specifications on Bankcard Interoperability V2.1*
- *EMV® QR Code Specification for Payment Systems (EMV QRCPS) Merchant-Presented Mode*
- *[ISO/IEC 18004] Information technology - Automatic identification and data capture techniques - QR Code bar code symbology specification*

- *[ISO 18245] Retail financial services -- Merchant category codes*
- *[ISO/IEC 8859-1] Information technology — 8-bit single-byte coded graphic character sets — Part 1: Latin alphabet No. 1*
- *[ISO/IEC 13239] Information technology — Telecommunications and information exchange between systems — High-level data link control (HDLC) procedures*
- *[ISO 3166-1 alpha 2] Codes for the representation of names of countries and their subdivisions - Part 1: Country codes, using two-letter country codes*
- *[ISO 4217] Codes for the representation of currencies and funds*
- *[ISO 7816-5] Identification cards — Integrated circuit cards — Part 5: Registration of application providers*
- *[ISO 639] Codes for the representation of names of languages – Part 1: Alpha - 2 Code*
- *[EMV Book 4] EMV Integrated Circuit Card Specifications for Payment Systems - Book 4 Cardholder, Attendant, and Acquirer Interface Requirements*
- *[Unicode] Unicode Standard, specifically the UTF-8 encoding form. For more information, please check: <http://www.unicode.org/versions/latest>*

Support

Please address your questions to the following service team:

Technical Specifications Team

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Word Convention

Convention	Tone	Implication
Shall	Mandatory	Must be followed without exception
Should	Optional but recommended	Required to follow but optional
Must	Mandatory	Required without exception
Must not	Mandatory	Not required
Could	Optional	Hypothetical
Can	Optional	Stating a capability
Cannot	Optional	Stating an incapability
May	Optional	Stating a possibility

Style Convention

Convention	Description
Boldface	Note/emphasis
<i>Italic</i>	Publications/book titles

1 Overview

The requirements for the QRC displayed by the Merchant, including its format and content, are described in the *EMV® QR Code Specification for Payment Systems (EMV QRCPS) Merchant-Presented Mode*. UPI merchant presented QRC is aligned with the EMV QRCPS with some additional requirements.

UnionPay International Technical Specification Team strongly recommends the member to read [EMV QRCPS] and the *UPI QR Code Payment Implementation Guide (Merchant Presented Mode) for Acquirers or Issuers* before reading this document.

2 QRC Payload and Formatting

2.1 Value Representation and Notation

For value representation of data objects, the following notation is used:

- Characters included in the Merchant-Presented QRC are enclosed in double quotation marks, for instance "Test@123".
- A character can be represented by its hexadecimal value. Single quotation marks are used to indicate the hexadecimal value, for instance '42' to represent the character "B".

For the presence of data objects, the following notation is used:

- M: Mandatory – shall always be present.

Note: If the Template, e.g., Merchant Information - Language Template, is present, then the data objects that are labelled M shall be present in the template. Otherwise, the whole template shall be absent.

- C: Conditional – shall be present under certain conditions.
- O: Optional – may be present.

2.2 Data Element Format Conventions

Table 1 Data Element Format Conventions

Data Element	Format Conventions
Numeric (N)	Values represented by all digits, so "0" to "9". The numeric alphabet includes ten (10) characters in total.
Alphanumeric Special (ans)	Values represented by the Common Character Set defined in [EMV Book 4]. The Alphanumeric Special alphabet includes ninety-six (96) characters in total (including the numeric alphabet and punctuation).
String (S)	Values represented by any character(s) defined in [Unicode]. A character in UTF-8 can be one to four byte long. The length must be equal to the number of characters in the value, not the number of bytes.

2.3 QRC Payload

The data elements and formatting are as follows.

Table 2 QRC Payload

ID	Name	Format	Length	Presence	Descriptions
"00"	Payload Indicator	N	"02"	M	Valid value: "01"
"01"	Point of Initiation Method	N	"02"	O	Valid values: "11" = Static QRC "12" = Dynamic QRC
"15"	Merchant Account Information	ans	Up to "99"	M	Compositions:

ID	Name		Format	Length	Presence	Descriptions												
						<table><tr><th>Data Field</th><th>Length</th><th>Format</th></tr><tr><td>Acquirer IIN</td><td>8</td><td>ans</td></tr><tr><td>Forwarding IIN</td><td>8</td><td>ans</td></tr><tr><td>Merchant ID</td><td>15</td><td>ans</td></tr></table> For example: “0001034400010344ABCDEFGHJKLMNOP”	Data Field	Length	Format	Acquirer IIN	8	ans	Forwarding IIN	8	ans	Merchant ID	15	ans
Data Field	Length	Format																
Acquirer IIN	8	ans																
Forwarding IIN	8	ans																
Merchant ID	15	ans																
“16”	Merchant Account Information	ans	Up to “99”	O	Reserved for UnionPay													
“02” - “14” & “17” - “25”	Merchant Account Information	ans	Up to “99”	O	Defined by other payment networks or Reserved by EMVCo													
“26” - “51”	Merchant Account Information	ans	Up to “99”	O	Templates reserved for additional payment networks.													
	“00” Globally Unique Identifier	ans	Up to “32”	M	The value is one of the following: • An Application Identifier (AID); • A UUID without the hyphen (-) separator; • A reverse domain name													
	“01”- “99” Payment network specific	S		O	Association of data objects to IDs and type of data object is specific to the Globally Unique Identifier.													
“52”	Merchant Category Code	N	“04”	M	The merchant category code must contain an MCC defined in the <i>UPI Operating Regulations</i> .													
“53”	Transaction Currency	N	“03”	M	The Transaction Currency must conform to <i>[ISO/IEC 4217]</i> and must contain the 3-digit numeric representation of the currency. For example, RMB is represented by “156”. The mobile													

ID	Name	Format	Length	Presence	Descriptions
					application should display the transaction currency in a readable way to the consumer, such as "RMB".
"54"	Transaction Amount	ans	Up to "13"	C	If the Transaction Amount (ID "54") is absent, the mobile application must prompt the consumer to enter the transaction amount. If the Transaction Amount (ID "54") is present, the mobile application must display the transaction amount with the transaction currency.
"55"	Tip or Convenience Indicator	N	"02"	O	Valid value: "01" = tip. If "01" is present, the mobile application must prompt the consumer to enter the tip. "02" = convenience fee fixed. If "02" is present, the mobile application must display the Value of Convenience Fee Fixed (ID "56") "03" = convenience fee percentage. If "03" is present, the mobile application must display the Value of Convenience Fee Percentage (ID "57")
"56"	Value of Convenience Fee Fixed	ans	Up to "13"	C	The Value of Convenience Fee Fixed must be present if the data object Tip or Convenience Indicator (ID "55") is present with a value of "02". Otherwise this data object must be absent. For example, if the Transaction Currency (ID "53") is "156" and the Value of Convenience Fee Fixed (ID "56") is "8.00", it indicates the merchant will charge 8 RMB convenience fee for the transaction.
"57"	Value of Convenience Fee Percentage	ans	Up to "05"	C	The Value of Convenience Fee Percentage must be present if the data object Tip or Convenience Indicator (ID "55") is present with a value of "03". Otherwise this data object must be absent.

ID	Name		Format	Length	Presence	Descriptions
						Valid value: "0.01"-"99.99". For example, "0.01" indicates that the convenience fee percentage is 0.01%, and the mobile application must calculate the convenience fee as Transaction Amount (ID "54" or entered by the consumer) * 0.01%.
"58"	Country Code		ans	"02"	M	The Country Code must contain a value defined by [ISO 3166-1 alpha 2] For example, China is represented by "CN".
"59"	Merchant Name		ans	Up to "25"	M	The Merchant Name (ID "59") can only use ASCII character set. If template "64" is present, the mobile application should check the consumer language preference with the merchant local language (ID "00" under template "64"). When they match, the mobile application should display the Merchant Name – Alternate Language (ID "01" under template "64"). Otherwise, the mobile application should display the Merchant Name (ID "59").
"60"	Merchant City		ans	Up to "15"	M	The Merchant City (ID "60") can only use ASCII character repertoire.
"61"	Postal Code		ans	Up to "10"	O	
"62"	Additional Data Field Template		S	Up to "99"	O	This template includes information that may be provided by the Merchant or may be populated by the mobile application to enable or facilitate certain use cases.
	"01"	Bill Number	ans	Up to "25"	O	If present, the content of the data object value for IDs "01" to "08" must be either "****" or a value defined by the merchant. The presence of "****" indicates that the mobile application should prompt the consumer to enter the information.
	"02"	Mobile Number	ans	Up to "25"	O	

ID	Name	Format	Length	Presence	Descriptions
"03"	Store Label	ans	Up to "25"	O	For a UPOP merchant, the reference label is used as the Order Number in the UPOP message.
"04"	Loyalty Number	ans	Up to "25"	O	
"05"	Reference Label	ans	Up to "25"	C Present if it is a UPOP merchant	
"06"	Customer Label	ans	Up to "25"	O	
"07"	Terminal Label	ans	Up to "25"	O	
"08"	Purpose of Transaction	ans	Up to "25"	O	Valid value: Any combination of "A", "M", "E" "A" = Address of the consumer. "M" = Mobile number of the consumer. "E" = Email address of the consumer. The data object with the ID "09" contains one or more values that indicate to the mobile application the data which must be provided as part of the transaction initiation. This data should already be known by the mobile application, and the consumer should not unnecessarily be prompted to input the data.
"09"	Additional Consumer Data Request	ans	Up to "03"	O	

ID	Name		Format	Length	Presence	Descriptions
	"10" -"49"	RFU for EMVCo	S		O	
	"50" -"99"	Reserved for payment network	S		O	
		"00" Globally Unique Identifier	ans	Up to "32"	M	The value is one of the following: - An Application Identifier (AID); - A UUID without the hyphen (-) separator; - A reverse domain name.
		"01"-"99" Payment System specific	S		O	Association of data objects to IDs and type of data object is specific to the Globally Unique Identifier.
"64"	Merchant Information - Language Template		S	Up to "99"	O	This template includes merchant information in an alternative language (not using the Common Character Set) and is provided as an alternative to the merchant information under the root. If the ID "64" is present, the Language Preference (ID "00") and Merchant Name (ID "01") must be present.
	"00"	Language Preference	ans	"02"	M	Language Preference should indicate the local language, and must contain a value defined by [ISO 639] For example, Chinese is represented by "zh".
	"01"	Merchant Name – Alternate Language	S	Up to "25"	M	The Merchant Name (ID "01" under template "64") should indicate the merchant name in the local language. For example, UnionPay International is represented by “银联国际” in Chinese. If template "64" is present, the mobile application should check

ID	Name	Format	Length	Presence	Descriptions
					the consumer language preference with the merchant local language (ID "00" under template "64"). When they match, the mobile application should display the Merchant Name – Alternate Language (ID "01" under template "64"). Otherwise, the mobile application should display the Merchant Name (ID "59").
	"02" Merchant City – Alternate Language	S	Up to "15"	O	The Merchant City (ID "02" under template "64") should indicate the merchant city in the local language. For example, Shanghai is represented by “上海” in Chinese.
	"03"- "99" RFU for EMVCo	S		O	
"65"- "79"	RFU for EMVCo	S	Up to "99"	O	Templates reserved for EMVCo
"80"- "99"	Unreserved Templates	S	Up to "99"	O	Unreserved Templates
	"00" Globally Unique Identifier	ans	Up to "32"	M	The value is one of the following: • An Application Identifier (AID); • A UUID without the hyphen (-) separator; • A reverse domain name
	"01"- "99" Context Specific Data	S		O	Association of data objects to IDs and type of data object is specific to the Globally Unique Identifier.
"63"	Cyclic Redundancy Check (CRC)	ans	"04"	M	The checksum must be calculated according to [ISO/IEC 13239] using the polynomial '1021' (hex) and initial value 'FFFF' (hex). The data over which the checksum is calculated must cover all data objects to be included in the QRC, including their IDs,

ID	Name	Format	Length	Presence	Descriptions
					Lengths and Values, in their respective order, as well as the ID and Length value of the CRC itself (but excluding its Value). Following the calculation of the checksum, the resulting two-byte hexadecimal value must be encoded as a 4 character Alphanumeric Special value by converting each nibble to an Alphanumeric Special character. Example: a CRC with a two-byte hexadecimal value of '007B' is included in the QRC as "6304007B".

3 Data Coding

For the Merchant-presented QR Code, all the data must be encoded using Byte Mode. If the Merchant-Presented QR Code includes characters other than Alphanumeric Special, then it must include an ECI mode indicator and ECI Designator, where the ECI Designator includes the binary representation of an ECI Assignment number equal to 000026, to indicate UTF-8 encoding.